

DR. ZAHRA MOKHTARI

Data Scientist | Statistical Physicist

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ABOUT ME

A data scientist with a PhD in statistical physics ([my peer-reviewed publications](#)), I build end-to-end solutions from data to deployment in production environments. My passion for finding patterns in data is backed by strong cross-disciplinary communication skills.

EXPERIENCE

PhD/Postdoctoral Researcher

University of Göttingen – Institute for Theoretical Physics

 2013 Oct. – 2019 Mar.  Göttingen, Germany

- Built parallelized simulations of stochastic models of bacterial migration, processing TB-scale trajectory data on HPC clusters.
- Managed cross-disciplinary collaboration (math/biology), to derive insights from simulations.
- Published first-author papers in peer-reviewed journals and presented results at international conferences.

Postdoctoral Researcher

Freie Universität Berlin (FU) – Department of Mathematics and Computer Science

 2020 Jul. – 2024 Nov.  Berlin, Germany

- Implemented agent-based simulations (C++/CUDA) to model pattern formation in colonies of ants, with up to ~400K agents
- Developed and applied image processing pipelines to quantify large-scale spatial patterns from simulation data.
- Mentored 50+ students in Physics, Math, Scientific Computing.
- Lectured on statistical mechanics and computational methods.

Parental Leave

 2019–2020, 2022–2023  Berlin, Germany

SKILLS

Languages Python SQL C/C++ ML & DL scikit-learn

TensorFlow Data Engineering & Tools Git Docker MLflow

Streamlit AWS GCP FastAPI Data Analysis

Statistical Analysis Matplotlib Tableau

LANGUAGES

Farsi: Native | English: Fluent | German: Upper-Intermediate (B2)

PROJECTS

Satellite Roof Segmentation

2026

- Binary semantic segmentation of rooftops from satellite imagery for solar panel planning.
- Compared deep learning architectures on a small/noisy dataset using transfer learning
- Prioritized recall over precision for real-world application needs

Unsupervised Music Pattern Discovery

 2025

 Le Wagon

- Applied unsupervised ML (K-means) to audio data for pattern extraction.
- Deployed as a Dockerized pipeline on Google Cloud; Streamlit visualization.

Agent-Based Modeling of Biological Patterns

 2022

 Freie Universität Berlin

- Built a GPU-accelerated simulation, with analysis in Python (NumPy/scikit-image).

EDUCATION

Certificate in Data Science and AI

Le Wagon

 Jan. 2025 – Mar. 2025  Berlin, Germany

- Hands-on ML, DL (DNNs, RNNs, LLMs), & engineering (MLflow, FastAPI).
- Pitched and led "Music Pattern Discovery" project (team of 4).

PhD in Statistical Physics

University of Göttingen

 2013 – 2018

 Göttingen, Germany

- Thesis – Bacteria in Heterogeneous Media

M.Sc. in Physics

Simon Fraser University

 2011 – 2013

 Vancouver, Canada

- Thesis – Quantum Ising chains in fields